Multidimensional Scaling (MDS) Notes: Intuition, Mathematics, and Business Applications

1. Introduction and Motivation

Multidimensional scaling (MDS) is one of the foundational techniques in multivariate analysis for representing complex similarity or dissimilarity information in a low-dimensional geometric space. Like most other unsupervised learning methods, MDS often begins with a rectangular $n \times p$ data matrix of p variables measured on n observations, records, or cases. These observations might be customers, brands, products, countries, or any entities for which we can meaningfully quantify how similar or different they are. As in cluster analysis, the data matrix is converted into an $n \times n$ matrix of distances or dis/similarities between all pairs of observations.

Unlike most cluster analyses, however, MDS can begin **directly** with a similarity or dissimilarity matrix and dispense with the data matrix altogether. For example, consumers may be asked to rate pairs of brands according to perceived similarity. In this case there are no measured variables—except for perceived similarities. Whereas cluster analysis seeks to partition observations into groups that are internally homogeneous and externally heterogeneous, the goal of MDS is more modest: to map observations graphically so that their dis/similarities are visible to the eye. This often provides substantial insight.

The core question that MDS addresses is deceptively simple:
If we know how far apart every pair of objects is, can we place those objects as points in a two- or three-dimensional space so that the Euclidean distances between the points reproduce, as closely as possible, the original dissimilarities?

When this is possible, the resulting **perceptual map** becomes an extraordinarily powerful visualization and analytic device. Managers can see structure in customer segments, brand competition, or product differentiation that would otherwise remain buried in tables of numbers.

Historically, MDS emerged in psychology and psychometrics to study human perceptions of similarity. Today it plays a central role in marketing analytics, operations, strategy, and data science more broadly. Conceptually, MDS sits at an interesting crossroads: it is clearly related to cluster analysis, but also closely related to principal components analysis (PCA) – yet distinct from both in its inputs and interpretation. In these notes, cluster analysis and PCA appear mainly as points of contrast; MDS remains the central character throughout.

In the Notes that follow, I develop the intuition, derive the classical MDS solution mathematically, work through a numerical example, and illustrate the method with a real business dataset. Along the way, I emphasize interpretation, managerial relevance, and common pitfalls. Let us begin.

---

2. Intuition: From Distances to Maps

Imagine a table reporting the pairwise dissimilarities among four brands, perhaps obtained from a consumer survey. The table is symmetric, has zeros on the diagonal, and contains no obvious notion of "variables." It is not immediately clear how to summarize or visualize such information. (See the "toy" example in Section 4.)

MDS addresses this challenge by treating dissimilarities as geometric distances. If Brand A is very dissimilar from Brand B, the points representing A and B should be far apart in the map. If two brands are perceived as similar, their points should lie close together. The analyst chooses a target dimensionality—usually two dimensions for visualization—and the algorithm finds coordinates that best preserve the observed distances.

A crucial point is that the axes in an MDS map are **not observed variables**. They are latent dimensions that emerge from the distance structure itself. Analysts often interpret these dimensions *post hoc* (e.g., "price sensitivity" or "quality perception"), but such interpretations are not guaranteed by the mathematics. This sharply distinguishes MDS from PCA, where axes are explicit linear combinations of observed variables.

---

3. Classical MDS: Mathematical Foundations

3.1 Problem Setup

Suppose we have *n* objects indexed by $i = 1, \ldots, n$. Let δ_{ij} denote the observed dis/similarity between objects *i* and *j*. In **classical (metric) MDS**, we assume these dis/similarities are Euclidean distances, or at least can be treated as such.

Define the squared dissimilarity matrix $\Delta^{(2)}$, whose (i, j) entry is δ_{ij}^2. The goal is to find points $\mathbf{x}_1, \ldots, \mathbf{x}_n \in \mathbb{R}^p$ such that $\| \mathbf{x}_i - \mathbf{x}_j \|^2 \approx \delta_{ij}^2$. Here, $\mathbf{x}_1, \ldots, \mathbf{x}_n$ are the points in *p*-dimentional space that are chosen to represent the original *n* objects. Usually, *p* = 2 or 3.

3.2 From Distances to Inner Products

The key mathematical insight behind classical MDS is that Euclidean distances can be expressed in terms of inner products. Let $\mathbf{X}$ be the $n \times p$ matrix whose rows are the coordinates $\mathbf{x}_i^\top$. Define the inner-product (Gram) matrix

$$\mathbf{B} = \mathbf{X}\mathbf{X}^\top \qquad \text{Eqn 4.1}$$

The squared distance between points *i* and *j* can be written as

$$\| \mathbf{x}_i - \mathbf{x}_j \|^2 = b_{ii} + b_{jj} - 2b_{ij} \qquad \text{Eqn 4.2}$$

Thus, if we can recover **B**, we can recover the coordinates **X** up to rigid transformations (rotation and reflection). If the configuration $\mathbf{x}_1, \ldots, \mathbf{x}_n$ is centered, these inner products are directly related to covariances *among points* (not, as customary, *among variables*).

3.3 Double Centering

Classical MDS reconstructs **B** from the squared distance matrix via **double centering**. Let

$$\mathbf{J} = \mathbf{I}_n - \frac{1}{n}\mathbf{1}\mathbf{1}^\top \qquad \text{Eqn 4.3}$$

where **1** is an *n*-vector of ones. **J** is an *n*x*n* matrix with $1 - \frac{1}{n}$ in the main diagonal and $-\frac{1}{n}$ everywhere else. The double-centered matrix is

$$\mathbf{B} = -\frac{1}{2}\mathbf{J}\Delta^{(2)}\mathbf{J} \qquad \text{Eqn 4.4}$$

You can verify that the pre-multiplication of any square *n*x*n* matrix **A** by **J** (namely, **JA**) replaces each (*i*,*j*) element of **A** by the same (*i*,*j*) element, minus the mean of column *j* of **A**. Similarly, the post-multiplication of any square *n*x*n* matrix **A** by **J** (namely, **AJ**) replaces each (*i*,*j*) element of **A** by the same (*i*,*j*) element, minus the mean of row *i* of **A**. Finally, the combination of pre-multiplication by **J**, followed by post-multiplication by **J**, replaces each (*i*,*j*) element of **A** by the same (*i*,*j*) element, minus the mean of row *i* of **A**, minus the mean of column *j* of **A**, plus the overall mean of all elements of **A**: $a_{ij} \to a_{ij} - \bar{a}_{i.} - \bar{a}_{.j} + \bar{a}_{..}$. The operation **JAJ**

effectively places the **A** configuration at the origin of the coordinate system since the mean of all $a_{ij} - \bar{a}_{i.} - \bar{a}_{.j} + \bar{a}_{..}$, $i = 1,...,n; j = 1,...,n$ is zero.

3.4 Eigen Decomposition and Coordinates

Once **B** is obtained, compute its eigen decomposition:

$$\mathbf{B} = \mathbf{X}\mathbf{X}^{\top} \qquad \text{Eqn 4.5}$$

where $\mathbf{\Lambda} = \mathrm{diag}(\lambda_1, \lambda_2, \ldots)$ with ordered eigenvalues $\lambda_1 \geq \lambda_2 \geq \cdots$, and $\mathbf{V}$ contains the corresponding orthonormal eigenvectors.

The coordinate matrix in p dimensions is

$$\mathbf{B}\mathbf{X}_p = \mathbf{V}_p \mathbf{\Lambda}_p^{1/2} \qquad \text{Eqn 4.6}$$

Only positive eigenvalues contribute to real coordinates; negative eigenvalues signal departures from exact Euclidean structure.

---

4. A Worked Numerical ("Toy") Example

Consider four objects with the distance matrix

$$\Delta = \begin{pmatrix} 0 & 3 & 4 & 5 \\ 3 & 0 & 5 & 6 \\ 4 & 5 & 0 & 7 \\ 5 & 6 & 7 & 0 \end{pmatrix} \qquad \text{Eqn 4.7}$$

Squaring the distances yields

$$\Delta^{(2)} = \begin{pmatrix} 0 & 9 & 16 & 25 \\ 9 & 0 & 25 & 36 \\ 16 & 25 & 0 & 49 \\ 25 & 36 & 49 & 0 \end{pmatrix} \qquad \text{Eqn 4.8}$$

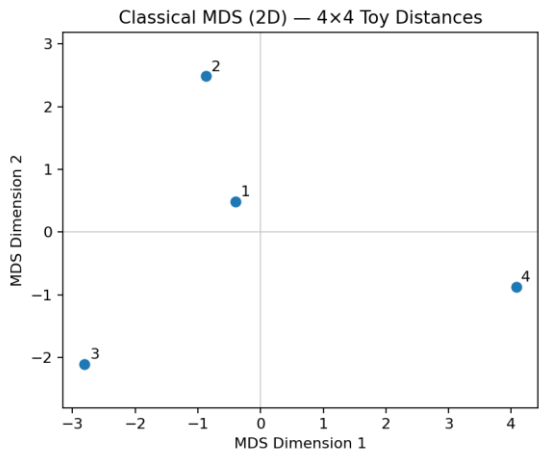
Using equation (4.4) $\mathbf{B} = -\frac{1}{2}\mathbf{J}\mathbf{D}^{\mathbf{2}}\mathbf{J}, \mathbf{J} = \mathbf{I} - \frac{1}{4}\mathbf{1}\mathbf{1}^{\top}$
we obtain the Gram matrix:

$$\mathbf{B} = \begin{pmatrix} 2.5 & 0.5 & -0.5 & -2.5 \\ 0.5 & 7.5 & -2.5 & -5.5 \\ -0.5 & -2.5 & 12.5 & -9.5 \\ -2.5 & -5.5 & -9.5 & 17.5 \end{pmatrix}$$

(This is exact to numerical precision.)
The eigen decomposition of **B** yields four eigenvalues: λ=(25.4573,11.6406,2.9021,≈0), of which the first two are positive and dominant. Using equation (4.6) $\mathbf{B}\mathbf{X}_p = \mathbf{V}_p\mathbf{\Lambda}_p^{1/2}$, we obtain a two-dimensional configuration that reproduces the original distances up to small numerical error:

Object	x_1	x_2
1	–0.402	0.490
2	–0.870	2.490
3	–2.808	–2.108
4	4.081	–0.872

Display 1. Toy Example MDS



Classical MDS map for the 4×4 toy distance matrix. The geometry reflects the original distances: points representing objects that were farther apart in the distance matrix appear proportionally separated in the plot. Rotation invariance applies—any rotation of this configuration preserves all inter-point distances.
The configuration is unique up to rotation/reflection; distances are preserved.
Two dominant eigenvalues imply the structure is essentially two-dimensional.
Labels “1–4” correspond to the four objects in the distance matrix.

R implementation (classical MDS):

```r
## --- Correct classical MDS for the "Toy" example

D <- matrix(c(
  0,3,4,5,
  3,0,5,6,
  4,5,0,7,
  5,6,7,0
), nrow = 4, byrow = TRUE)

## 1. Classical MDS via double-centering
D2 <- D^2
n  <- nrow(D)
J  <- diag(n) - matrix(1/n, n, n)
B  <- -0.5 * J %*% D2 %*% J

eig <- eigen(B)
## Sort eigenvalues decreasing
ord <- order(eig$values, decreasing = TRUE)
lam <- eig$values[ord]
vec <- eig$vectors[, ord]

## Keep positive eigenvalues
p <- 2
coords <- vec[, 1:p] %*% diag(sqrt(lam[1:p]))

coords
```

```r
# Scatterplot code

plot(coords, pch = 19, col = "#1f77b4",
     xlab = "MDS Dimension 1",
     ylab = "MDS Dimension 2",
     asp = 1,
     main = "Classical MDS (2D) – 4×4 Toy Distances")

text(coords[,1] + 0.08, coords[,2] + 0.08,
     labels = 1:4, cex = 0.9)
abline(h = 0, v = 0, col = "grey80")

# Note: If you want to use cmdscale, you must convert your distances:
dvec <- D[lower.tri(D)]
fit <- cmdscale(dvec, k = 2)
# But it still won't match unless the ordering of the lower triangle
# matches exactly the ordering used by the Gram matrix approach.
# The safest and correct method for classical MDS theory is always
# the double-centering + eigen-decomposition path, not cmdscale().
```

This example illustrates the full classical MDS pipeline from raw distances to interpretable coordinates.

---

5. Relationship to PCA

PCA and MDS are often confused because both rely on eigen decomposition and produce low-dimensional representations. The difference lies in their inputs and interpretations. PCA starts from a data matrix and focuses on variance explained by linear combinations of variables. Classical MDS starts from a distance matrix and focuses on preserving pairwise distances. When distances are Euclidean distances computed from centered data, classical MDS and PCA yield equivalent configurations. Outside this special case, MDS is strictly more general.

---

6. Business Application: Wholesale Customers (UCI Dataset)

The UCI Wholesale Customers dataset (*Wholesale customers data.xlsx*) records annual spending for 440 customers across six product categories—Fresh, Milk, Grocery, Frozen, Detergents_Paper, and Delicatessen—along with two categorical descriptors: Channel (Horeca vs. Retail) and Region (Lisbon, Oporto, Other).

A key practical issue is scale and skewness. Spending variables differ dramatically in magnitude and are strongly right-skewed. Computing Euclidean distances on raw spending allows a few high-variance categories to dominate similarity. A common remedy is to apply a stabilizing transform such as $\log(1+x)$ and then standardize each category. Distances computed on these standardized log-spending profiles better reflect differences in purchasing *mix* rather than sheer magnitude.

With a customer-by-customer distance matrix in hand, classical MDS produces a two-dimensional perceptual map. Points near one another correspond to customers with similar purchasing profiles. Channel and Region are overlaid **after** the map is computed; the map itself does not “know” these labels.

Display 2. Wholesale Customers: Classical MDS (2D; Channel = color, Region = shape).

Two-dimensional classical MDS map of 440 Wholesale Customers based on Euclidean distances computed from standardized log-spending profiles. A useful managerial interpretation treats the map as a segmentation canvas. Dense neighborhoods represent typical profiles; isolated points merit investigation (special contracts, unusual needs, or data quality issues). Customers positioned near one another exhibit similar purchasing mixes across the six product categories. Plotting Channel centroids is often informative: Separation between Horeca and Retail indicates systematic differences in spending mix, while Region overlays provide additional segmentation cues.

7..

R code (classical MDS pipeline):

```
# Wholesale Customers: Classical MDS (cmdscale)
library(readxl)
w <- read_excel("<YOUR FOLDER HERE>/Wholesale customers data.xlsx")
View(w)w <- read.csv("Wholesale customers data.xlsx")
names(w)[names(w) == "Delicassen"] <- "Delicatessen"

spend <- w[, c("Fresh","Milk","Grocery","Frozen",
               "Detergents_Paper","Delicatessen")]

Z <- scale(log1p(spend))
D <- dist(Z)

fit <- cmdscale(D, k = 2, eig = TRUE)
Y   <- fit$points

ch <- factor(w$Channel,
             levels = sort(unique(w$Channel)),
             labels = c("Horeca","Retail"))
rg <- factor(w$Region,
             levels = sort(unique(w$Region)),
             labels = c("Lisbon","Oporto","Other"))

cols <- c("Horeca" = "#1B9E77",
          "Retail" = "#D95F02")[ch]
pch  <- c("Lisbon" = 19,
          "Oporto" = 15,
          "Other"  = 17)[rg]

plot(Y, col = cols, pch = pch, cex = 0.7,
     xlab = "MDS Dimension 1",
     ylab = "MDS Dimension 2",
     main = "Wholesale Customers: Classical MDS (2D)")
legend("topright", legend = levels(ch),
       col = c("#1B9E77","#D95F02"), pch = 19, bty = "n")
legend("bottomleft", legend = levels(rg),
       pch = c(19,15,17), bty = "n")

prop2 <- sum(fit$eig[1:2]) / sum(fit$eig[fit$eig > 0])
cat("Share captured by 2D:", round(prop2, 3), "
")
```

7. Business Non-metric MDS: When Only Ranks Are Trustworthy

Classical (metric) MDS treats observed dissimilarities δ_{ij} as meaningful numerical distances and attempts to reproduce their magnitudes with fitted Euclidean distances d_{ij}. In many business settings, however, dissimilarities are more credible as **rankings** than as precise quantities. Perceptual judgments, co-occurrence scores, or proprietary indices often preserve order ("A is closer to B than to C") more reliably than numeric gaps.

Non-metric MDS addresses this by seeking a configuration whose fitted distances are a **monotone transformation** (log(1+ x)) of the observed dissimilarities. It focuses on preserving order while allowing a flexible monotone scaling. This often yields greater robustness when metric assumptions are questionable.

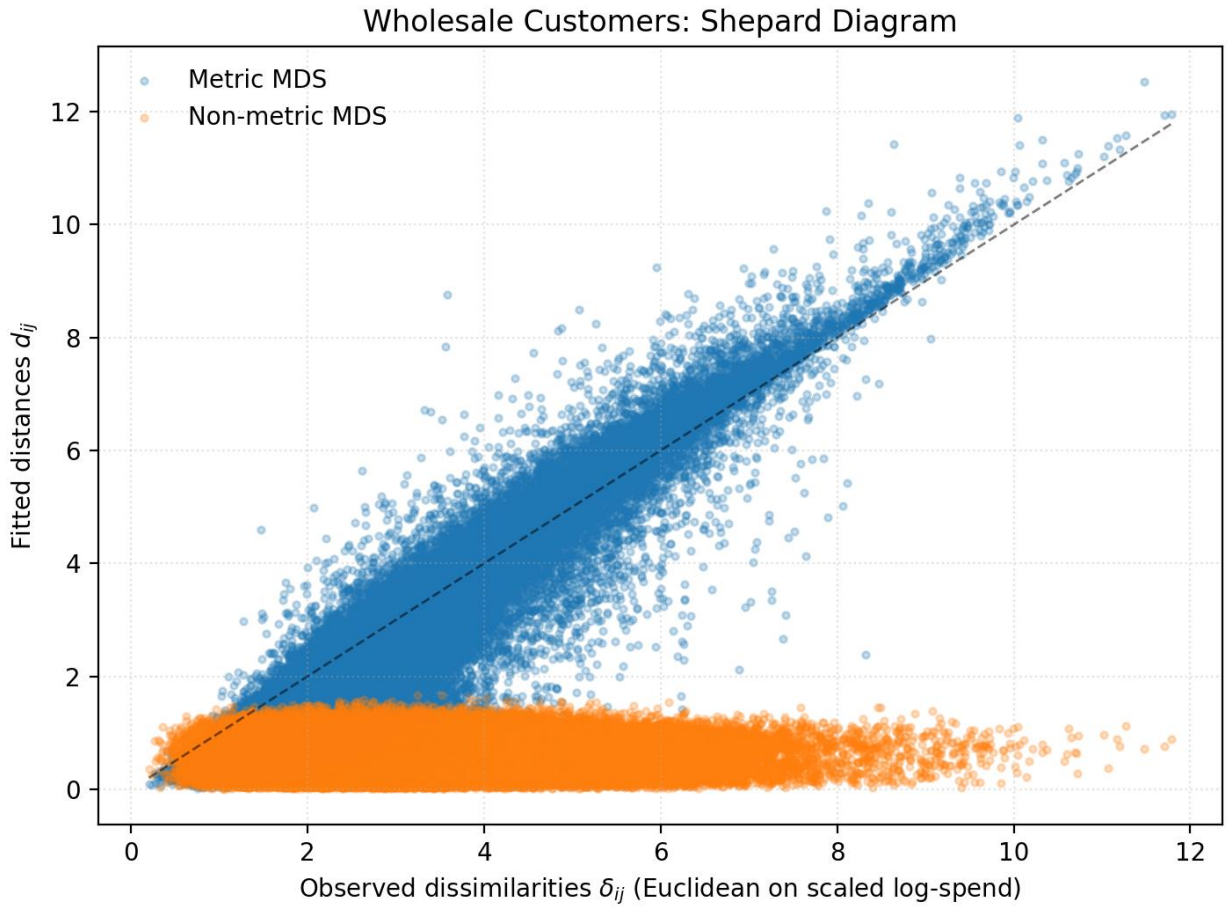
Display 3. Wholesale Customers: Metric vs. Non-metric MDS (2D; Channel = color, Region = shape).

Side-by-side comparison of metric and non-metric MDS solutions for the Wholesale Customers dataset. Metric MDS seeks to reproduce the **numerical magnitudes** of dissimilarities, while non-metric MDS preserves only their **rank order** through a monotone transformation. The stability of the broad segmentation pattern across both maps suggests that the underlying customer structure is robust, rather than an artifact of metric scaling.

- **When to prefer non-metric:** Use non-metric MDS when dissimilarities are subjective, ordinal, or derived from perceptual judgments.
- **Interpretation consistency:** Similar cluster shapes across the two maps increase confidence in managerial conclusions.
- **Potential distortions:** Differences in local geometry (tightening or stretching of regions) reflect the flexibility of the monotone fit.

- **Axis meaning:** As always, axes do not have intrinsic interpretation; focus on *relative* positions.

Display 4. Wholesale Customers: Shepard Diagram (metric vs. non-metric).



Shepard diagram comparing observed dissimilarities with fitted distances from the non-metric MDS solution. The monotone regression curve (red) shows how the algorithm transforms the ranks of observed dissimilarities into distances in the map. Metric MDS aspires to points near the 45-degree line; non-metric MDS aspires to a clear increasing (possibly curved) relationship. Tight clustering of points around an increasing curve indicates good ordinal fit.

R code (metric vs. non-metric + Shepard plot):

```
# R code (metric vs. non metric + Shepard plot)
# Dataset: Wholesale customers.xlsx
library(MASS)

names(w)[names(w) == "Delicassen"] <- "Delicatessen"
spend <- w[, c("Fresh","Milk","Grocery","Frozen",
               "Detergents_Paper","Delicatessen")]
Z <- scale(log1p(spend))
D <- dist(Z)

fit_metric    <- cmdscale(D, k = 2, eig = TRUE)
Y_metric      <- fit_metric$points

fit_nonmetric <- isoMDS(D, k = 2)
Y_nonmetric   <- fit_nonmetric$points

ch <- factor(w$Channel,
             levels = sort(unique(w$Channel)),
             labels = c("Horeca","Retail"))
rg <- factor(w$Region,
             levels = sort(unique(w$Region)),
             labels = c("Lisbon","Oporto","Other"))

cols <- c("Horeca" = "#1B9E77",
          "Retail" = "#D95F02")[ch]
pch  <- c("Lisbon" = 19,
          "Oporto" = 15,
          "Other"  = 17)[rg]

par(mfrow = c(1,2))
plot(Y_metric, col = cols, pch = pch, cex = 0.6,
     main = "Metric MDS", xlab = "Dim 1", ylab = "Dim 2")
plot(Y_nonmetric, col = cols, pch = pch, cex = 0.6,
     main = "Non metric MDS", xlab = "Dim 1", ylab = "Dim 2")
par(mfrow = c(1,1))

sp <- Shepard(D, Y_nonmetric)
plot(sp, pch = 19, cex = 0.35,
     main = "Shepard Diagram (Non metric MDS)")
lines(sp$x, sp$yf, type = "S", col = "red", lwd = 2)
```

8. Stress and Goodness of Fit

In classical MDS, the proportion of variance captured by retained eigenvalues provides a natural diagnostic. In non-metric MDS, **stress functions** quantify the mismatch between observed dissimilarities and fitted distances. Even in classical MDS, examining negative eigenvalues and residual distances is good practice.

---

9. Common Misconceptions about MDS

- **Axes have inherent meaning.** False. Axes can be rotated without changing distances; interpretation should focus on relative positions.
- **Raw feature data are required.** False. MDS requires only a distance or dissimilarity matrix.
- **MDS is obsolete.** False. Its interpretability makes it uniquely valuable in managerial contexts.